SAT MATH

Practice tests • Practice Test 1 • Module 1 (22 questions, 35 minutes) • Answer key

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Answer key with worked solutions

Directions

These questions follow the format of the SAT Math section. A calculator is allowed on every question (Desmos is built into the digital SAT). Multiple-choice questions have one best answer. For **student-produced responses**, write your answer on the line: a fraction or a decimal is accepted (for example $3/4$ or 0.75); give negative answers with a minus sign; do not use mixed numbers.

Question 1 (Algebra)

If $3x + 8 = 29$, what is the value of x?

(A) 7

(B) $\dfrac{37}{3}$

(C) 21

(D) 3

Solution

Answer: (A). $3x = 21$, $x = 7$.

Question 2 (Algebra)

The function f is defined by $f(x) = 5x - 2$. What is the value of $f(3)$?

(A) 15

(B) 17

(C) 13

(D) 1

Solution

Answer: (C). $5(3) - 2 = 13$.

Question 3 (Data Analysis)

A recipe uses flour and sugar in the ratio $3 : 2$. How many cups of sugar are needed with 12 cups of flour?

(A) 6

(B) 8

(C) 18

(D) 10

Solution

Answer: (B). $\frac{3}{2} = \frac{12}{s}$, so $s = 8$.

Question 4 (Advanced Math)

Which expression is equivalent to $(x+3)^2$?

(A) x^2+9
(B) x^2+3x+9
(C) x^2+6x+6
(D) x^2+6x+9

Solution
Answer: (D). $(x+3)(x+3) = x^2+6x+9$.

Question 5 (Algebra) STUDENT-PRODUCED RESPONSE

What is the solution to $2(x-4) = x+5$?

Solution
Answer: 13. $2x-8 = x+5$, $x = 13$.

Question 6 (Geometry and Trig)

A rectangle is 12 cm long and 5 cm wide. What is the length, in cm, of its diagonal?

(A) 17
(B) 13
(C) $\sqrt{119}$
(D) 60

Solution
Answer: (B). $\sqrt{12^2+5^2} = \sqrt{169} = 13$.

Question 7 (Advanced Math)

What are the solutions of $x^2-9x+20 = 0$?

(A) 4 and 5
(B) -4 and -5
(C) 2 and 10
(D) -2 and -10

Solution
Answer: (A). $(x-4)(x-5) = 0$.

Question 8 (Algebra)

What is the slope of the line through $(1,4)$ and $(3,10)$?

(A) $\frac{1}{3}$
(B) 2
(C) 6
(D) 3

Solution
Answer: (D). $\frac{10-4}{3-1} = 3$.

Question 9 (Data Analysis)

A jacket costs \$80 before a 15% sales tax. What is the cost including tax?

(A) \$95
(B) \$68
(C) \$92
(D) \$81.20

Solution
Answer: (C). $80(1.15) = 92$.

Question 10 (Algebra)

If $x + 2y = 11$ and $x - y = 2$, what is the value of x?

(A) 5
(B) 3
(C) 7
(D) 4

Solution
Answer: (A). Subtract: $3y = 9$, $y = 3$, so $x = 5$.

Question 11 (Advanced Math) STUDENT-PRODUCED RESPONSE

The function f is defined by $f(x) = x^2 - 3x + 2$. What is the value of $f(-2)$?

Solution
Answer: 12. $4 + 6 + 2 = 12$.

Question 12 (Geometry and Trig)

What is the area of a circle with radius 6?

(A) 12π
(B) 36π
(C) 6π
(D) 144π

Solution
Answer: (B). $\pi r^2 = 36\pi$.

Question 13 (Algebra)

Which describes all solutions of $x - 7 \geq 15$?

(A) $x \geq 8$
(B) $x \leq 22$
(C) $x \geq -8$
(D) $x \geq 22$

Solution
Answer: (D). Add 7: $x \geq 22$.

Question 14 (Data Analysis)

What is the mean of the data set 4, 7, 9, 12, 18?

(A) 9
(B) 10
(C) 12.5
(D) 14

Solution
Answer: (B). $\frac{50}{5} = 10$. (The median is 9.)

Question 15 (Advanced Math)

A town's population is modeled by $P = 2000 \cdot 2^{t/5}$, where t is the number of years after 2020. Which statement is true?

(A) The population doubles every 5 years.
(B) The population increases by 2000 each year.
(C) The population doubles every year.
(D) The population increases by 5% each year.

Solution
Answer: (A). When t increases by 5, the exponent $t/5$ increases by 1, so P is multiplied by 2.

Question 16 (Algebra) STUDENT-PRODUCED RESPONSE

For the linear function g, $g(0) = 4$ and $g(2) = 10$. What is the value of $g(5)$?

Solution
Answer: 19. Slope $\frac{10-4}{2} = 3$, so $g(x) = 3x + 4$ and $g(5) = 19$.

Question 17 (Advanced Math)

What is the solution to $\sqrt{x+7} = 4$?

(A) -3
(B) 23
(C) 9
(D) 16

Solution
Answer: (C). $x + 7 = 16$, $x = 9$. Check: $\sqrt{16} = 4$.

Question 18 (Geometry and Trig)

In a right triangle, one angle measures 30° and the hypotenuse is 10. What is the length of the side opposite the 30° angle?

(A) $5\sqrt{3}$
(B) $10\sqrt{3}$
(C) 20
(D) 5

Solution
Answer: (D). $10\sin 30^\circ = 5$.

Question 19 (Advanced Math) STUDENT-PRODUCED RESPONSE

What is the sum of the solutions of $x^2 - 7x + 10 = 0$?

Solution
Answer: 7. $(x-2)(x-5) = 0$: $2+5=7$.

Question 20 (Data Analysis)

What is 45% of 200?

(A) 45
(B) 90
(C) 155
(D) 110

Solution
Answer: (B). $0.45(200) = 90$.

Question 21 (Algebra)

What is the y-intercept of the graph of $5x + 3y = 30$?

(A) $(0, 10)$
(B) $(6, 0)$
(C) $(0, 6)$
(D) $(0, 30)$

Solution
Answer: (A). $x = 0$: $3y = 30$, $y = 10$.

Question 22 (Advanced Math) STUDENT-PRODUCED RESPONSE

If $x^2 = 49$ and $x < 0$, what is the value of x^3?

Solution
Answer: -343. $x = -7$, so $x^3 = -343$.